MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 1102: INTRODUCTION TO PROGRAMMING WITH C

END OF SEMESTER EXAMINATION

DATE: AUGUST 2022 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

i. There are **5** questions in this paper. **ANSWER ALL.**

ii. There are **THREE** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

QUESTION 1

a. Differentiate the following:

i. REPEAT UNTIL and WHILE DO loops. **[2 Marks]**

ii. IF THEN and IF THEN ELSE statements. **[2 Marks]**

iii. Compiler and Interpreter. **[2 Marks]**

iv. 1GL and 2GL. **[2 Marks]**

v. Structured and Unstructured programming. **[2 Marks]**

b. Write a pseudocode for a program that prints numbers from 1 to 20 in reverse order. **[4 Marks]**

c. Write a pseudocode for a program that calculates and prints the square of a number if it is even and the cube of a number if it is odd. The numbers ranges from 5 to 30. Use a DO WHILE loop. ***Note: Do not use the modulus.*** **[5 Marks]**

QUESTION 2

a. Write a C program that implements a factorial of a number as follows: *For any given positive number n, n! = n(n-1)*(n-2)*.....*1. For instance: 4! = 4*3*2*1 = 24.* **[5 Marks]**

b. Explain the use of **#include<string.h>** as used in C. **[2 Marks]**

c. Use a while loop, write a program in C that prints a π^n where n is from 2 to 6 inclusively and π is 3.14 which is to be declared as a symbolic constant. **[4 Marks]**

d. Write a program in C that solves the following equation: $\mathbf{2^k \cos(k) - \sqrt{a^k + b^2}}$ where the user will enter k, a *and* b as positive integers. **[5 Marks]**

e. Apart from the position of the unary operator, explain the other difference bettween a postfix decrement and prefix decrement with a valid example. **[4 Marks]**

QUESTION 3

a. Write an algorithm in a form of a flowchart for a program that asks a user to enter a series numbers on the keyboard. The entry of the numbers should be terminated by entering a 999. The program should count and print the number of even numbers from the series. **[6 Marks]**

b. Write a C function that implememnts a **"void"** keyword. **[4 Marks]**

c. Using any loop construct, write a program in C that will output multiplication tables of 1 to 10. **[6 Marks]**

d. Can you use **"int"** data type to store **32769** value? Explain your answer. **[2 Marks]**

QUESTION 4

a. Write a program that will implement a simple quiz game. The program should present a user with three mathematical questions *(you should decide the questions to use)*. For each question, a user should be given four trials to get it right otherwise the program should jump to the next question automatically after he/she fails. The program should track the correct answers and give a score at the end. To make it interesting, for the correct answers, the program should tell the user the number of trials that were made to get the question right. The user should be given a chance to play the game again or to exit. **[10 Marks]**

b. Write a program that will output the following: **[5 Marks]**

1	**2**	**3**	**4**	**5**
1	**4**	**9**	**16**	**25**
1	**16**	**81**	**256**	**625**

c. Write a program implemented using user-defined functions to find the minimum and maximum number among three numbers entered by a user. The functions should be Max and Min respectively. *The functions should get three numbers as parameters.* **[7 Marks]**

d. With an example, explain how a **typedef** is used in C programming. **[4 Marks]**

e. Explain **three** reasons why the study of algorithms and data structures is of paramaount importance in computer science. **[6 Marks]**

QUESTION 5

a. Show **two** ways on how you can declare a constant variable in C. **[2 Marks]**

b. Mention **two** reasons why it is important to use user defined functions in a program. **[4 Marks]**

c. What will be the output given the following code snippet: **[4 Marks]**

```
for (int x=2; x<=3; x++){printf("The value is %d\n", x);if(x%2==0)printf("Hey programmer");}printf("Program has terminated.");
```

d. With a valid example, explain the difference between a postfix decrement and prefix increment apart from the position of the unary operator. **[4 Marks]**

END OF QUESTION PAPER